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identification news

on the move

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PHILIPS

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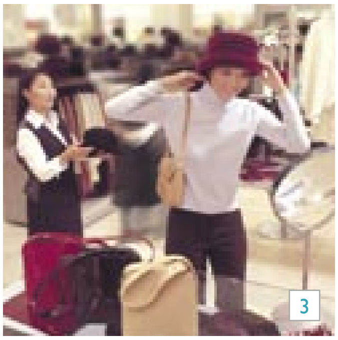


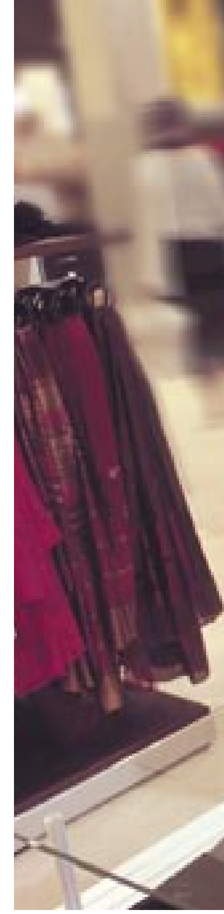
Brand protection:
countering the counterfeiters

The figures tell an alarming story. Today's market for counterfeit goods is worth USD 250 billion. Twenty-two percent of all clothing and footwear sold worldwide is counterfeit, and designer fakes have cost the industry 100,000 jobs[1]. With figures like this, it's not surprising leading fashion labels, manufacturers and retailers are urgently looking for new ways to beat the counterfeiters and protect valuable brands.

Business lost to counterfeiting is estimated to represent up to 8 % of world trade. What's more, the rate of growth exceeds that of legitimate worldwide trade. For brand owners, there's the immediate financial loss. Longer term, counterfeiting can destroy brand names, ruin reputations, even drive companies out of business.

RFID provides extra weapon against piracy

Against this background, RFID (Radio Frequency Identification) provides an important weapon in the fight against counterfeiting. It's particularly relevant in the fashion industry. However, no sector is immune from piracy or the need for protection. Brand name companies >



and manufacturers of all kinds of high-value goods, from liquor to jewelry, are seeking ways to secure their products.

RFID can take a key role because it enables easy, instant recognition of products. Every product can be labeled with an IC with a unique ID or serial number (UID). Simply pass the item in front of a reader and you can check if it's the real thing. In addition, complete boxes of products can be checked without opening them.

"Today, fashion house representatives visit stores to check how their products are being displayed and sold. In future, they may bring along RFID readers to check the authenticity of the products as well," says Rainer Lutz, Segment Marketing Manager Fashion & Retail Market Sector RFID, Philips Semiconductors. "If products are labeled with RFID tags, it's easy to assure shelf availability and supply sales data, because this can be electronically collected at the point-of-sale."

Tailored security

RFID offers greater versatility as well. ICs and antennas can be built into a variety of label types, from paper hang-tags and care labels to reusable hard tags with integrated Electronic Article Surveillance (EAS) features. Manufacturers can then tailor the type and level of protection to the value of their goods. While low-value items might be tagged with labels that support just a UID code, the technology allows for many more options, such as those offered by Philips ICODE ICs.

For instance, manufacturers might want to add a lockable 'signature' to a read/writeable tag. The signature can also be encrypted for extra security. It's even possible to implement 'secure memory management' on the RFID chip, so only authorized users can access the data.

"At Philips, we've been looking into fighting counterfeiting for many years," adds Rainer. "In fact, we received an award for 'Outstanding Achievement in RFID for product protection' at PISEC 05, a major international conference for the brand protection, product authentication, document security and RFID industries. And we're working with top fashion houses and sportswear companies who want to adopt RFID to protect their garments."

No diversions

Of course, the advantages aren't just in brand protection and verification. RFID brings benefits right from the manufacturing stage. A single tunnel reader can count 60 shirts in a box within one second, and throughout the supply chain, RFID adds speed and visibility.

This brings another valuable advantage – it helps clamp down on parallel and grey markets. Companies can use RFID to keep a constant eye on their goods, so they can spot instantly if they're diverted into different channels. And if they do turn up in unexpected places, if the RFID label is still attached, it's easy to identify the source and intended destination.

Authenticity guaranteed

Traceability and the ability to check the source of goods is also why airplane makers, airlines and the automotive industry are among the leading adopters of RFID. After all, when it comes to safety-critical parts, the last thing anyone wants is to discover they are using a potentially inferior and even dangerous counterfeit.

> "If products are labeled with RFID tags, it's easy to assure shelf availability and supply sales data."

Traceability is also critical in pharmaceuticals, and of course food. Many national governments insist on producers and retailers enabling 'field to fork' tracking. In Europe, Regulation (EC) 178/2002 also requires that producers be able to recall products within 48 hours if a problem is detected.

Cutting the cost of recalls

Obviously, a recall is a costly matter. It's also a logistical challenge, and can adversely impact a company's brand >

reputation. Here too, RFID helps, particularly if tagging is at item level, because manufacturers can quickly recall specific batches of product, rather than complete stocks. It's faster, less expensive and offers consumers rapid reassurance.

Enhanced consumer benefits

What about the consumer? Clearly, when we buy a brand name product, we expect it to be just that – especially if it's a high price-ticket product. Similarly, everyone wants to be sure that products such as foods, pharmaceuticals or safety-critical vehicle parts are from a known and trusted source. So brand protection is in everyone's interest.

Another exciting aspect of RFID is its potential to extend brand protection into an enhanced shopping experience. Imagine a fashion house wants to offer its customers an additional 'thank you' for buying its products. It could embed extra data in an appropriate RFID chip. Then consumers could simply wave their NFC (Near Field Communication) enabled cellphone across the RFID label and instantly receive a gift voucher on their phone, or an invitation to a fashion show or some other loyalty incentive.

It's an attractive way to discretely encourage consumers to check the authenticity of products (because if the label wasn't valid, they'd be alerted), while simultaneously encouraging brand loyalty. Of course, customers need to be clearly informed about the use of RFID labels and should always have the choice of having the label removed or killed at checkout.

Affordable protection, with all-round advantages

While many companies have seen the advantages of RFID in logistics, their security departments have been looking elsewhere for anti-counterfeit solutions. Today, that is changing. The same RFID technology and infrastructures that companies are using to improve visibility and efficiency in their supply chain is now also providing a powerful, affordable weapon against counterfeiting and parallel markets.

Link to: www.philips.semiconductors.com/applications/smart_cards/index.html

[1] Source: US Customs Service (RL PPT July 2005)

Counterfeit drugs can kill

Most of us are aware of the existence of counterfeit products - from Rolex watches to designer outfits, CDs to DVDs. However, few of us would ever consider that the medicines in our local pharmacy or drugstore might also be counterfeit.

The World Health Organization (WHO) estimates that 10 % of medicines in the $550 billion international drug market are counterfeit. Lifestyle drugs, such as impotence medicines Viagra or Cialis, and obesity drugs such as Reductil are most at risk. However, the list of the top 30 most counterfeited drugs worldwide includes almost all anti-retrovirals used to treat Aids, many antibiotics, as well as expensive drugs used to treat cancer and infertility.

The numbers are worse in developing countries, often with more serious consequences. The WHO estimates that up to 25 % of medicines traded in these developing countries are fake. In Lagos, Nigeria, 80 % of drugs in one market were bogus. The Chinese Government attributes 192,000 deaths a year to fake medicines. In Cambodia, 70 % of the malarial drug artensuate is counterfeit. Globally, the WHO estimates 200,000 deaths a year from malaria could be avoided if the medicines were real.

Fortunately, RFID offers the same benefits to the Pharmaceutical industry as it does in the retail sector – guaranteeing authenticity, reducing parallel and grey market trading. Going further, the pharmaceutical industry is also planning to tag individual prescriptions. At the item level, HF smart labels are better suited than UHF labels, as they can be more easily read even when densely packed amongst liquid-filled medicine bottles or metal-foil pill packages. As a result, the EPCglobal HLS BAG has decided to support HF labels for pharmaceutical item level tagging.

A number of real world trials have already taken place, and some medical giants have announced plans to tag individual prescriptions with RFID labels by the end of this year. With an estimated $12 billion annual loss to counterfeit pharmaceuticals, the pharmaceutical industry is serious about implementing RFID. And for patients, guaranteeing they get the right medicine could be a matter of life and death.

Source: WHO website

Germany to issue smart passports

Governments all over the world are planning to bring in smart passports, aiming to reduce forgery, raise security levels for travelers and speed up border control procedures. One of the first to move from planning to implementation, Germany has chosen smart passport chips from Philips, the only volume supplier capable of delivering the right ICs – in the right quantities.

"Biometrics makes travel safer and simpler."

As you may recall from the previous issue of On The Move, Philips is working closely with governments around the world as they plan to introduce smart passports. It's a move that's being driven by the United States' Visa Waiver Program (VWP), which requires visitors to hold a machine-readable passport when visiting the country for less than 90 days.

Germany is one of the first countries to start issuing smart passports. Bundesdruckerei GmbH, the leading German passport manufacturer, has chosen Philips' 72 Kbytes EEPROM Smart Passport chip from its Smart*MX* secure smart card IC family. This family of ICs exceeds the specifications for smart passports set by the International Civil Aviation Organization (ICAO). It is also the first to achieve Common Criteria EAL5+ certification, the highest level of security certification for secure contactless smart card solutions.

In addition to the highly secure 72 Kbytes EEPROM, dual interface (ISO/IEC7816, ISO/IEC14443) PKI IC, the Smart*MX* family also includes a 36 Kbytes Smart Passport chip and a triple interface (ISO/IEC7816, ISO/IEC14443 and USB) version that meets the upcoming data security and memory requirements of eGovernment projects.

Otto Schily, Germany's Minister of Internal Affairs said while presenting the project on June 1st, "Biometrics makes travel safer and simpler – and the roll-out of passports with biometric data in Europe is an important element in the fight against organized crime and international terrorism. On 9 January 2002, the German government passed a new anti-terrorism law which included the use of biometrics in passports. As forgers are becoming more and more creative it's imperative to use the latest techniques. We need to build on existing high standards and raise the security levels of identity documents throughout Europe". [1]

Philips is working with a variety of partners to implement the German smart passport scheme. T-Systems is supplying the operating system and the on-chip passport application, and Sokymat GmbH is providing the chip inlay.

For more on the Smart*MX* family, visit:
www.semiconductors.philips.com/markets/identification/products/smartmx

'The smart passport prepares for take-off', On the Move 7.2:
www.philips.semiconductors.com/news/publications/index.html#onthemove

[1] Source: Bundesministerium PR e-pass 1 June 2005

Football supporters score with NFC technology

Roda JC is making history. It's the first football club in the Netherlands to trial NFC technology in mobile phones. During this season, 50 Roda JC fans will use NFC-enabled phones that offer a lot more convenience and features than their traditional Club Card. The project is a close collaboration between the forward-thinking club, Dutch football association KNVB, Dutch telecommunications provider KPN, Philips, Bell ID and SmartPoint. Success with this innovative 'mobile ticketing' will mark the beginning of a new era in consumer-oriented services and create major potential for marketing. >

During the current 2005 – 2006 season, the 50 Roda JC fans taking part in the trial will use their phones to access the Parkstad Limburg stadium. The special mobile phones (Nokia 3220s) are equipped with an NFC chip which takes over all functions of the Club Card. As well as access to the stadium, fans can also buy goods from food stands and the supporters shop using the phones. Roda JC uses a closed payment system within the stadium, so contactless payment with card or phone is the only way to purchase items.

The supporters participating in the trial can load credit onto the NFC chip by holding their phones up to the credit-loading machine, and can easily check their balance on the phone's screen. In later stages of the trial it will be possible to purchase match tickets using the phone. Supporters will also be able to load credit onto their phones from their mobile networks, making the credit-loading machine redundant.

Frank Rutten from Roda JC said, "We started the trial on 28 August, giving NFC-enabled phones to 50 of our season card-holders representing a cross-section of our supporters. The trial will show us how well mobile ticketing is accepted and what challenges we may face for a full-scale implementation. Of course it is too early to draw conclusions, but initial feedback from the fans is very positive."

"The club enjoys being a trailblazer of new ideas," Frank continued, "and we are very enthusiastic about an advanced technology that brings so many potential advantages. Our fans share this enthusiasm too and feel proud that their stadium is at the forefront of this new technology. On top of benefits such as convenience, a key factor about NFC on phones is that it allows us to have a new level of interaction with our supporters, for example sending them player details during a match."

The KNVB is watching the trial closely. If successful they hope it will be adopted by other clubs and eventually by the entire premier division. "Mobile ticketing using NFC offers clubs and supporters considerable added-value, especially as extra functionality is added," said Wilfred de Rooij, KNVB. "If the technology is then adopted at other stadiums the benefits increase even further, with every club able to provide a personalized, direct service."

As the trial progresses the KNVB, Roda JC and KPN will be looking at whether other services on the phones would be of interest to the club's supporters. These may include personal services such as news and loyalty programs, sent as text messages or via mobile internet.

Frank Zandee of KPN: "Now people can use their phones to communicate with other devices as well as other people. Couple this to the fact that mobile phones have screens for displaying information – from the remaining balance on their e-purse to company logos – and it presents major marketing opportunities. Also, as NFC emulates existing MIFARE implementations it immediately opens up the available market."

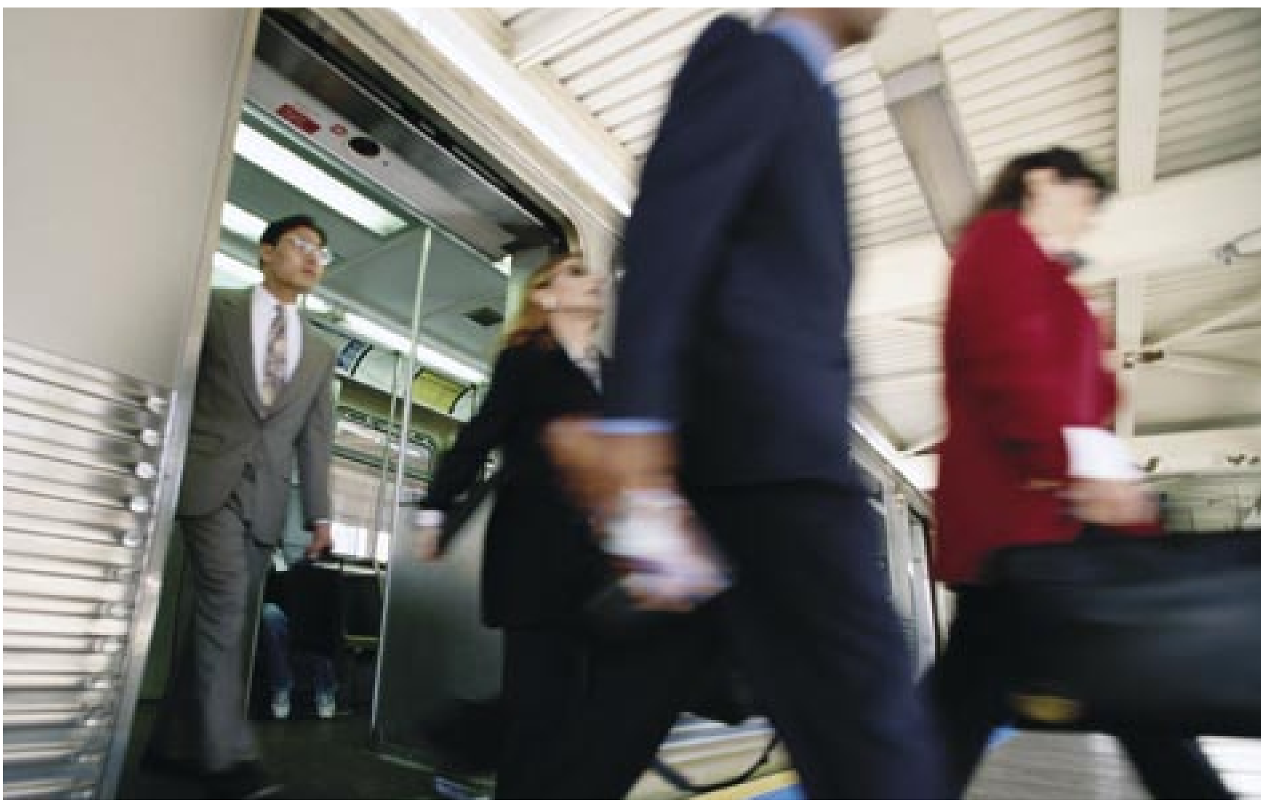
"Mobile ticketing using NFC offers clubs and supporters considerable added-value."

"For KPN, the trial has been an eye-opener to the complex value chain of the ID industry. We have learned so much just in working with everyone and setting up the pilot. We suspect that supporters, indeed consumers, will enjoy the freedom and extra services offered by NFC mobile – which will open the flood gates of opportunity."

For more information on Philips' NFC mobile solutions, visit: www.philips.semiconductors.com/applications/smart_cards/nfc_mobile/

European transport schemes adopt multi-application cards

Widely implemented in many large scale transport networks in Asia, contactless ticketing solutions are now being increasingly adopted throughout Europe. Many European cities and countries are installing contactless ticketing systems, often looking beyond their transport networks by implementing solutions capable of supporting additional applications such as contactless payment, club memberships and loyalty schemes.



We already reported on pilot transport ticketing projects for systems being deployed in Oslo, London and Translink the country-wide project in the Netherlands. Now we look at two further large-scale implementations using Philips' MIFARE® DESFire smart cards, in the city of Turku in Finland and for Czech Railways.

Turku set for city-wide DESFire solution

In implementing a multi-application smart card network, Turku, Finland is the first city in Scandinavia and Europe to adopt a city-wide solution using the MIFARE DESFire smart card. As well as making the city's public transport easier to experience and use, the DESFire solution delivers the power, flexibility and high security needed to support additional applications such as payment services. It also replaces the existing proprietary card system with one that is fully ISO 14443A compliant – the de facto standard for transport ticketing – although both systems will initially run in parallel.

With over 20 million passengers using public transportation in Turku every year, the new electronic >

“Our customers will benefit from faster transactions.”

ticketing will give users hassle-free entry to their chosen mode of transport. The cards also can be configured easily for different customer groups. For example, the Turku speciality fifty-fifty card, which combines a season card with economy tickets to provide regular travellers with the best value.

However, the real benefit of the new system is that once it is established in the public transportation network, the DESFire smart cards can be extended to incorporate additional services. For example, they could also act as a membership card to allow access to leisure facilities such as sports halls and swimming pools.

The system integrator responsible for installing the new system is Buscom Oy, and the availability of both Philips and third party supporting tools helped facilitate the integration of DESFire into Buscom’s transport terminals. Buscom’s COO, Riku Niemelä said, “The long-lasting, excellent relationship between Philips and Buscom led to an efficient operation, from the specification phase of DESFire through to the smooth roll-out of the Turku system. Buscom had full access to early samples and best-in-class technical support.”

Multi-application cards for Czech Railways

Handling almost 180 million passengers in 2004, Czech Railways is committed to ensuring its customers have maximum convenience and security. To this end, a major pilot scheme using a new national transport card called ‘In-karta’ is being launched. In-karta also uses the MIFARE DESFire platform and passengers will initially use their smart cards solely as a transport pass. Later it will be extended to include an electronic wallet, allowing passengers to buy tickets, newspapers and refreshments, not only at Czech Railways but also at bus transport operators.

For stage one of the roll-out, cards will be issued to employees of Czech Railways and their families, and also to pensioners. In total some 200,000 cards will be used to test the technology. This will rise to nearly one million cards, for Czech Railways alone, when the project goes live. The card will serve as travel pass, service card and catering card as well as an ID card for railway employees.

In stage two, the project will be expanded to include contractors. Card functionality will also be expanded by adding e-purse and loyalty-card applications, making it a truly multi-application card. The system will then be opened to the general public later this year.

“We are going to overtake competitors with this unique project and show our ability to defend our position as the Czech Republic’s most important transport operator,” said Tomáš Čech, Vice-President of Passenger Transport Department, Czech Railways. “Our customers will benefit from faster transactions for a number of goods and services and the simplicity of using a single card for them all.”

A choice solution for transport networks

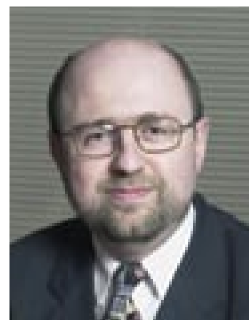
Offering a flexible and secure solution, Philips’ MIFARE DESFire is the obvious choice for multi-application implementations. It features 4 Kbytes of non-volatile memory, a high speed triple DES data encryption co-processor and flexible memory organisation structure. Guaranteeing data integrity in contactless transactions, MIFARE DESFire uses a mutual 3-pass authentication technique, together with a true random number generator and an anti-tear mechanism. It is fully compliant with existing ISO 14443A contactless ticketing systems, as well as being open to any further applications our partners may want to add – right from the start.

For more information on the MIFARE product family, visit:
/markets/identification/products/mifare/index.html

"The demand for cards and readers for eGovernment applications is expected to grow dramatically in the coming years."

Partner perspective

Welcome to our new Partner perspective, where our partners air their views on the Identification market. In this first article, Kurt Schmid, CEO of OMNIKEY, gives his views on the smart card business, both present and future.



Kurt Schmid

OMNIKEY

Headquartered in Walluf, Germany, and with offices and R&D facilities at five other international locations, OMNIKEY develops, manufactures and implements smart card based read / write devices, as well as biometric and contactless (RFID) smart card readers for the corporate 'smart identity' market. With sales of over a million readers per year, it is one of the leading companies in this market and specializes in developing readers for any application regardless of computer or operating system. Part of the ASSA ABLOY Identification Technology Group (ITG), OMNIKEY's sister companies include HID, Indala and Sokymat, also well-known names in the identification industry.

"In the ID market," said Kurt, "experience and reliability counts. OMNIKEY has a vast accumulated know-how in all aspects of the smart card industry; I myself have been involved since it began over 20 years ago. We have built up a large installed base, ensuring proven reader solutions that deliver seamless operation. Our understanding of key standards has been vital to our success – because it translates into a reliable product, one that is not only compliant with standards but also interoperable with the broad range of smart cards and applications in the industry.

Standards and interoperability

"Our involvement with developing standards 'at source' helps enormously. As an associate member of the PC/SC Workgroup, OMNIKEY contributed the key part covering support for contactless smart cards within the new PC/SC 2.0 specification. Together with companies such as Philips and Microsoft, we helped define this interface standard to enable robust communication between a PC and a contactless smart card. Our goal is to ensure any reader will work in any application and accept any type of card. It is this interoperability which will help drive the take up of contactless cards in the smart identity market. An open standard also helps the market to grow as it allows anyone to develop new applications. Of course, all of OMNIKEY's reader products are ready for PC/SC 2.0.

"Our association with Philips goes beyond the development of standards. We began working together with contact readers in 2000 and now we are investing in contactless solutions, and looking into creating

>

NFC readers. Our PC know-how is complemented by Philips' silicon solutions and knowledge of emerging markets. Together, our combined competences put us in prime position to explore and take advantage of new applications.

Smart identity market

"We primarily use Philips' MIFARE reader ICs (CL RC632) in our contactless readers. These ICs feature ISO 14443A and B interfaces, as well as ISO 15693 – one of the standards commonly used in physical access control systems. This is one part of the smart identity market and one of our current target areas with corporate customers. The other area is network security, involving single sign-on and digital signatures for EPN / VPN (Electronic / Virtual Private Network) systems, where secure recognition of the person logged on to the computer is essential.

"Presently, many customers in the corporate sector are using two separate identification solutions – one for physical access control and one for network log-in. Often, cards are 'dual chip' solutions incorporating separate contact and RF components. However, with IT divisions increasingly handling both logical security and physical access, the market is now moving towards a single-chip, dual-interface solution. Industry watchers are predicting that as much as a third of all smart identity installations in the next few years will be fully integrated physical and logical access solutions.

Smart passports

"Beyond this, there is an increasing trend towards contactless only solutions. A major factor in driving this shift is the take-up of contactless solutions in smart passports. Their large scale adoption in eGovernment systems such as health cards, national ID and smart passports is increasing public awareness of the capabilities of contactless cards for secure identification applications.

"The demand for cards and readers for eGovernment applications is expected to grow dramatically in the coming years. To be able to grow with this market, OMNIKEY needs a major, committed partner capable of high volume silicon supply. Philips fits the bill perfectly. By partnering companies who are also serious about the ID market, OMNIKEY can maintain its leadership position."

Single solution for contactless MasterCard and Visa card

Speeding checkout times and delivering increased consumer convenience, contactless payment is becoming increasingly popular - particularly in the USA. Increasing demand for contactless cards for these applications means card manufacturers want a simpler, quicker way to deliver products that satisfy the different needs of their customers. Answering this need, Philips' MIFARE® CASH delivers a single Cash Alternative Smart Chip solution that meets the contactless payment specifications for Visa and MasterCard, as well as being fully-compliant with the ISO 14443A contactless standard.

Based on Philips' proven contactless smart card technology, MIFARE CASH is tailored to meet the commercial constraints of high volume, single application payment schemes. Featuring embedded software optimized for the payment market, it delivers fast transaction speeds at a low unit cost.

Card manufacturers can configure either the MasterCard Paypass or the Visa MSD payment application during pre-personalization. Simplifying integration, the pre-packaged chip is supplied in the well-established MOA4 contactless module, which has been qualified by the card manufacturing industry. Application approval is expected at the beginning of 2006, when the product will be available to all qualified manufacturers of MasterCard and Visa payment cards.

Reinforcing Philips' leadership in contactless smart card technology, MIFARE CASH demonstrates the company's continued support for the smart card industry as it gears up for the rapidly emerging contactless payment market.

Banking: www.semiconductors.philips.com/applications/smart_cards/banking/banking/
MIFARE: www.semiconductors.philips.com/markets/identification/products/mifare/

Identification updates

Semiconductor manufacturers team up to accelerate UCODE EPC Gen2 deployment

Philips and Texas Instruments are cooperating on conformance testing for the technical interpretation of the EPCglobal IncTM Electronic Product CodeTM (EPC) Generation 2 RFID standard. Their goal is to ensure worldwide interoperability, faster market deployment and multiple sourcing for Gen 2 products.

Philips and TI have worked together in the past to enable and accelerate the adoption of international RFID standards, such as the high-frequency RFID standard ISO/IEC 15693.

Read more: www.philips.semiconductors.com/news/content/file_1171.html

More awards for London's PRESTIGE fare collection system

PRESTIGE, Transport for London's regional automatic fare collection system, has added two more major awards to its collection: Best Operational Transport Project, and the Grand Prix as the Best Operational Project – all sectors. Judges from distinguished companies and organizations selected PRESTIGE as the best Private Finance Initiative (PFI): "PRESTIGE is an outstanding project that has delivered real improvements for the benefit of the community it serves. It represents a triumph in a very difficult environment and has become a benchmark for AFC systems throughout the world."

Dutch Government tightens security with Philips

As of autumn 2005, the Dutch Government buildings in The Hague will be equipped with a contactless access control system based on Philips' RFID technology. The system's primary goal is to tighten security, but the 2,000 access cards will also help international visitors find foreign newspapers at the digital newspaper kiosk. And they'll allow people to identify themselves for billing purposes – at the café and restaurant, for example.

Philips and HP drive adoption of UCODE EPC Gen2 RFID solutions

Working together, Philips and HP aim to accelerate adoption of the global EPC Class 1 Generation2 (Gen2) RFID standard. The collaboration brings together the complementary expertise of the two companies, allowing them to affect the whole RFID value chain. They will provide optimized Gen2 technology, solutions and services for migrating from previous RFID standards. Ensuring seamless migration and global interoperability, Gen2 RFID helps solve supply chain challenges in key areas such as CPG / retail, hi-tech and pharmaceuticals.

Read more: www.philips.semiconductors.com/news/infocus/gen2_RFID/

Meet us at the following events

Event:	**ID World**
Date:	2 - 4 November, 2005
Location:	Rome, Italy
Website:	www.idworldonline.com

Event:	**CarteS**
Date:	15 - 17 November 2005
Location:	Paris, France
Website:	www.semiconductors.philips.com/cartes2005

Event:	**CES**
Date:	5 - 8 January, 2006
Location:	Las Vegas, USA
Website:	www.cesweb.org

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